

Skill week – 6

Session - 11

Task 1-command history

Store Commands, Maintain History Buffer, Handle Capacity Limits, Retrieve Entries, Display History, Validate Consistency

```
ganesh@DESKTOP-79N3772: ~  
1366 pwd  
1367 echo "Current Directory: $PWD"  
1368 OLDPWD="$PWD"  
1369 cd /tmp  
1370 pwd  
1371 cd "$OLDPWD"  
1372 pwd  
1373 touch temp_test.txt  
1374 ls temp_test.txt  
1375 rm temp_test.txt  
1376 ls temp_test.txt  
1377 true  
1378 echo $?  
1379 false  
1380 echo $?  
1381 MYNAME="UbuntuStudent"  
1382 echo "$MYNAME"  
1383 export MYNAME  
1384 printenv MYNAME  
1385 export COURSE="OperatingSystems"  
1386 echo "$COURSE"  
1387 printenv MYNAME  
1388 printenv COURSE  
1389 export COURSE="LinuxShell"  
1390 echo "$COURSE"  
1391 bash -c 'echo "Child process: $MYNAME"'  
1392 bash -c 'echo "Child process course: $COURSE"'  
1393 export 123ABC=hello  
1394 history
```

History

```
ganesh@DESKTOP-79N3772:~$ echo "Command 1"  
Command 1  
ganesh@DESKTOP-79N3772:~$ echo "Command 2"  
Command 2  
ganesh@DESKTOP-79N3772:~$ echo "Command 3"  
Command 3  
ganesh@DESKTOP-79N3772:~$ echo "Command 4"  
Command 4  
ganesh@DESKTOP-79N3772:~$
```

Store some commands

```
ganesh@DESKTOP-79N3772:~$ history 10
1390 echo "$COURSE"
1391 bash -c 'echo "Child process: $MYNAME"'
1392 bash -c 'echo "Child process course: $COURSE"'
1393 export 123ABC=hello
1394 history
1395 echo "Command 1"
1396 echo "Command 2"
1397 echo "Command 3"
1398 echo "Command 4"
1399 history 10
ganesh@DESKTOP-79N3772:~$
```

Display recent history

```
ganesh@DESKTOP-79N3772:~$ history | tail -5
1396 echo "Command 2"
1397 echo "Command 3"
1398 echo "Command 4"
1399 history 10
1400 history | tail -5
ganesh@DESKTOP-79N3772:~$ !120
-bash: !120: event not found
ganesh@DESKTOP-79N3772:~$ 120 echo "Command 1"
120: command not found
ganesh@DESKTOP-79N3772:~$
```

Retrieve a history entry

```
ganesh@DESKTOP-79N3772:~$ echo $HISTSIZE
1000
ganesh@DESKTOP-79N3772:~$ echo $HISTFILESIZE
2000
ganesh@DESKTOP-79N3772:~$ history -w
ganesh@DESKTOP-79N3772:~$ history -r
ganesh@DESKTOP-79N3772:~$ history 10
2397 echo "Command 2"
2398 echo "Command 3"
2399 echo "Command 4"
2400 history 10
2401 history | tail -5
2402 120 echo "Command 1"
2403 echo $HISTSIZE
2404 echo $HISTFILESIZE
2405 history -w
2406 history 10
ganesh@DESKTOP-79N3772:~$
```

Check history configuration/capacity

Validate history consistency

Task 1 – Observation:

Commands were successfully stored and retrieved using the history buffer. History entries, capacity settings, and consistency were successfully verified.

TASK 2 — Pipeline Structures

```
ganesh@DESKTOP-79N3772:~$ echo "Hello Linux" | wc -w
2
ganesh@DESKTOP-79N3772:~$
```

Simple pipeline

```
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\norange\n" | sort | uniq
apple
banana
orange
ganesh@DESKTOP-79N3772:~$
```

Pipeline with three commands

```
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\norange\n" | sort | uniq
apple
banana
orange
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\norange\n" | sort | uniq | wc -l
3
ganesh@DESKTOP-79N3772:~$
```

This demonstrates a **multi-process pipeline**.

```
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\norange\n" | sort | uniq | wc -l
3
ganesh@DESKTOP-79N3772:~$ ls | grep ".txt"
destination.txt
sample.txt
source.txt
ganesh@DESKTOP-79N3772:~$ printf "a.txt\nb.c\nc.txt\nd.c\n" | grep ".txt"
a.txt
c.txt
ganesh@DESKTOP-79N3772:~$
```

This sends the output of `ls` to `grep`.

This sends the output of `ls` to `grep`.

If there are no `.txt` files, it may produce no output — that's okay.

```
C.txt
ganesh@DESKTOP-79N3772:~$ printf "3\n1\n2\n" | sort
1
2
3
ganesh@DESKTOP-79N3772:~$
```

Test execution order

```
ganesh@DESKTOP-79N3772:~$ echo "Pipeline: printf -> sort -> uniq -> wc"
Pipeline: printf -> sort -> uniq -> wc
ganesh@DESKTOP-79N3772:~$ printf "cat\ndog\ncat\nbird\n" | sort | uniq | wc -l
3
ganesh@DESKTOP-79N3772:~$
```

Validate pipeline layout

Task 2 – Observation:

Pipeline commands were successfully created and executed by connecting multiple processes. Execution order, process communication, and different pipeline configurations were successfully tested.

Session – 12

Task 1

Create Pipes, Redirect Streams, Execute Connected Commands, Close Descriptors, Synchronize Processes, Test Data Flow

```
ganesh@DESKTOP-79N3772:~$ echo "Hello from process 1" | cat
Hello from process 1
ganesh@DESKTOP-79N3772:~$
```

Basic pipe

```
ganesh@DESKTOP-79N3772:~$ echo "Hello Linux" | wc -w
2
ganesh@DESKTOP-79N3772:~$
```

Pipe with wc

```
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\norange\n" | grep apple
apple
apple
ganesh@DESKTOP-79N3772:~$
```

Pipe with grep

```
ganesh@DESKTOP-79N3772:~$ printf "3\n1\n4\n2\n" | sort
1
2
3
4
ganesh@DESKTOP-79N3772:~$
```

pipe with sort

```
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\norange\n" | sort | uniq
apple
banana
orange
ganesh@DESKTOP-79N3772:~$
```

multiple connected commands

```
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\norange\n" | sort | uniq | wc -l
3
ganesh@DESKTOP-79N3772:~$
```

test complete dataflow

printf → sort → uniq → wc it demonstrates

```
ganesh@DESKTOP-79N3772:~$ echo "Start" && echo "Process completed"
Start
Process completed
ganesh@DESKTOP-79N3772:~$
```

Test data flow

Observation – Task 1

Observation:

Pipes were successfully created and used to connect commands. Data was transferred between processes, and different pipeline configurations were successfully tested.

TASK 2 — Multiple Pipes

Now we need to demonstrate **multiple pipes and multiple processes**.

```
ganesh@DESKTOP-79N3772:~$ printf "5\n2\n8\n1\n5\n" | sort | uniq
1
2
5
8
ganesh@DESKTOP-79N3772:~$
```

Two pipes

```
ganesh@DESKTOP-79N3772:~$ printf "5\n2\n8\n1\n5\n" | sort | uniq
1
2
5
8
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\norange\nbanana\n" | sort | uniq | wc -l
3
ganesh@DESKTOP-79N3772:~$ printf "cat\ndog\ncat\nbird\ndog\n" | sort | uniq | tr 'a-z' 'A-Z' | wc -l
3
ganesh@DESKTOP-79N3772:~$ printf "red\nblue\nred\ngreen\nblue\nyellow\n" | sort | uniq | grep -v green | wc -l
3
ganesh@DESKTOP-79N3772:~$ printf "apple\nbanana\napple\ncherry\nbanana\norange\n" | sort | uniq | tr 'a-z' 'A-Z' | grep
"A" | wc -l
3
ganesh@DESKTOP-79N3772:~$
```

Testing three and a long pipe

Observation – Task 2

Observation:

Multiple pipes were successfully used to connect several processes. Long pipelines were executed in the correct order, and data flow between processes was successfully verified.